A Three-Stage PPO Curriculum for the Continuous Art Gallery Problem

Document 2 of 2: Saturate → Prune → Refine

AGP Project — Design Note

September 20, 2026

Abstract

Document 1 defined a single MDP for the Art Gallery Problem in which coverage and guard parsimony enter as two potential-based reward components. This document specifies the training curriculum on top of it. Stage SATURATE (SATURATE) switches the guard cost off entirely and asks only for coverage, with a boundary-inward placement prior that is annealed away; the policy learns to fill the gallery, deliberately over-completely. Stage PRUNE (PRUNE) unmasks a delete action, switches the guard cost on, and pays a bounded bonus for every removal that costs no coverage, decaying smoothly as coverage is lost. Stage REFINE (REFINE) replaces the fixed regularity penalties with state-wise constraints and polishes on the hardest instances. The design borrows four specific devices from the humanoid locomotion literature — one reward basis with a per-stage weight column, a task objective held at exactly zero weight until its prerequisite is learned, scaffolding terms deleted rather than decayed, and multi-critic advantage weighting — and contributes three things those papers do not provide: a quantitative stage-transition gate, a blended switch with a shadow-trained parsimony critic, and a cheap privileged *prune oracle* that can warm-start the delete action by imitation. We give the full weight schedule, PPO hyperparameters per stage, network architecture, the instance-difficulty curriculum, an ablation protocol, and a concrete implementation roadmap against the existing code.

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1 Why stage the training at all

1.1 The conflict

The AGP objective has two parts that pull against each other during *learning*, though not at the optimum. Coverage is easy to improve from a random start — almost any guard placed in an empty gallery captures a large fraction of the area — and its gradient is dense and informative. Parsimony is the opposite: a guard-count penalty is meaningful only once the agent can reliably cover the gallery, because before that point the cheapest way to reduce the guard count is to stop placing guards, which also destroys coverage. Turning both on from step zero means the agent must discover, from a random policy, that the *only* route to a high return runs through a temporarily expensive regime.

This is the same structure Wang et al. [20] describe for humanoid dribbling:

> "Training a humanoid robot to dribble presents a fundamental challenge: the rewards associated with walking and dribbling often conflict and are difficult to balance. Walking requires the robot to maintain stability and move efficiently, while dribbling demands agility and precise ball control. This trade-off complicates the simultaneous optimization of both skills."

Their ablation quantifies the cost of ignoring it: a single-stage policy reaches 33.0% success on dribble-to-target and 0% on obstacle avoidance, against 86.7% and 92.3% for the two-stage curriculum. The failure mode they report — "one-stage training policy can't stably walk and

Table 1: The three locomotion references, compared on the dimensions that matter here. Local copies are in `references/locomotion/`.

Paper	Stages	What changes at a stage boundary	Transition trigger
Dribble Master [20]	2	Task reward $0.0 \to 1.5$; approach proxy $2.0 \to 1.0$; imitation scaffold $1.2 \to 0.0$; ball spawn $10\,\text{m} \to 2\,\text{m}$; camera FOV $2\times \to 1\times$ real	Qualitative only: "once the robot consistently demonstrates reliable locomotion"
Vision-driven soccer [19]	1 (explicitly single-stage)	— (no curriculum; uses potential-based shaping and a multi-critic split instead)	n/a
Agile striker [21]	4	Task reward $0.0 \to 4.0$; tracking proxy $2.0 \to 0.5$; gait scaffolds $3.0/1.0 \to 0.0$; conflicting regulariser (feet yaw) $-1.0 \to 0.0$; spawn annulus $[0.5, 4] \to [0.5, 2]\,\text{m}$; then DAgger distillation; then constrained online RL	Qualitative only: "once the teacher policy acquires a basic kicking skill"

control the ball simultaneously, as both capabilities are learned jointly from scratch" — is exactly the failure mode we expect from an AGP policy trained with the guard cost on from the beginning.

1.2 What the three reference papers actually do

Table 1 summarises the three papers, with an eye to what is reusable.

Four patterns are worth importing wholesale.

P1. One reward basis, one weight column per stage. Neither multi-stage paper changes the observation space, the action space or the reward *terms* between stages — only the coefficients (and the initial-state distribution). We adopt this exactly: table 2 is the entire curriculum.

P2. The objective you cannot yet optimise gets weight zero, not a small weight. Dribble Master holds ball-velocity tracking at 0.0 through stage 1; the striker paper holds its ball-velocity reward at 0.0 through stage 1. Neither uses a small-but-nonzero coefficient. Our guard cost is therefore *exactly* 0 in SATURATE — which is also what was asked for in the project brief.

P3. Scaffolds are deleted; proxies are demoted. Both papers zero their imitation/gait scaffolding at the boundary (Dribble Master: reference joint position $1.2 \to 0.0$, "removed to promote exploration and discovery of effective dribbling behaviors"; striker: feet swing $3.0 \to 0.0$, arm swing $1.0 \to 0.0$), but keep the stage-1 proxy alive at a reduced weight ($2.0 \to 1.0$ and $2.0 \to 0.5$ respectively) rather than removing it, as insurance against catastrophic forgetting. Document 1's *anneal-iff-non-potential* rule explains *why* this split is the right one: scaffolds are non-potential and bias the optimum, proxies closer to the objective do not.

P4. Split the critic before you split the reward. Wang et al. [19] found that "using a single critic to estimate combined rewards can cause negative interference between distinct reward components," and use $A_{\text{total}} = 2A_{\text{goal}} + 1 \cdot A_{\text{aux}}$. Our coverage and parsimony rewards differ by more than an order of magnitude in scale and by a factor of ~ 30 in per-step variance, so this applies with force.

1.3 What we have to design ourselves

None of the three papers gives a quantitative criterion for advancing a stage. Dribble Master says "once the robot consistently demonstrates reliable locomotion and target pursuit"; the striker paper says "once the teacher policy acquires a basic kicking skill". There is no threshold, no step count, and no automated gate anywhere. Nor does any of them address the value-function discontinuity that a hard reward switch creates. Section 7 supplies both. A third gap — how to get a policy to *use* a newly unmasked action that it has never sampled — is addressed in section 4.2 with a privileged prune oracle, which is the AGP analogue of the teacher–student step in Xu et al. [21] but far cheaper, because unlike a robot teacher our oracle is a twenty-line greedy loop.

2 Curriculum overview

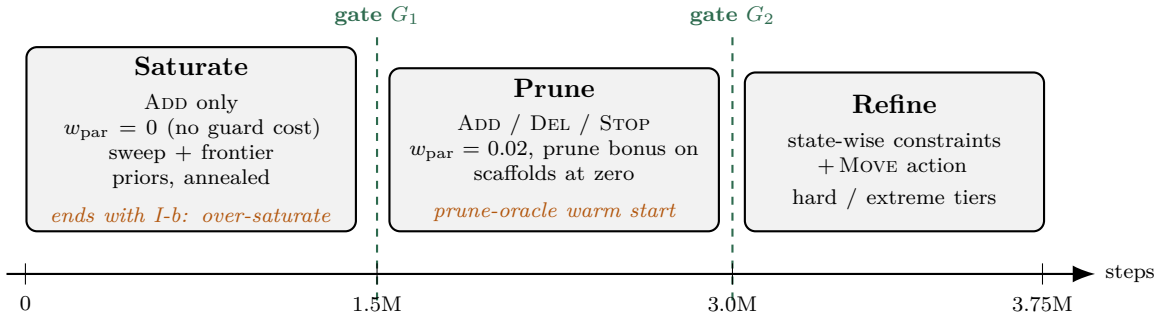


Figure 1: The curriculum. Step counts are targets, not fixed schedules: each boundary is crossed when its performance gate (section 7) fires, with the step count acting only as a fallback cap. Weights are linearly interpolated across a 50k-step blend window rather than switched instantaneously.

3 Stage I — Saturate

3.1 Objective and configuration

SATURATE answers one question: *can the policy cover an arbitrary gallery at all?* The guard cost is zero, so the telescoped episode objective (Document 1, the *telescoping proposition*) reduces to

$$R_{\text{SATURATE}}(\tau) \;=\; w_{\text{cov}}\, c_T \;+\; w_{\text{goal}} \mathbf{1}\left[c_T \geq c^{\dagger}\right] \;-\; w_{\text{fail}} \mathbf{1}\left[c_T < c^{\dagger}\right] \;+\; (\text{scaffolding}),$$

i.e. pure maximum coverage. This is a problem with a known, strong, cheap baseline: greedy maximum coverage, which inherits the $1-1/e$ guarantee from submodularity [11]. SATURATE is not expected to beat greedy. It is expected to *match* greedy while (a) using continuous positions rather than a candidate set and (b) producing, as a side effect, exactly the over-complete guard sets that PRUNE needs.

Action mask:
mode restricted to {ADD}. This makes the action space identical to the current $[0,1]^2$ implementation.

Instance distribution $\mathcal{D}$:
all four difficulty tiers, but sampled with a difficulty-annealed weighting (section 10): $(0.55, 0.30, 0.10, 0.05)$ over (easy, mid, hard, extreme) at step 0, linearly reaching uniform by the end of the stage.

Termination:
at $c_t \geq c^{\dagger}$, on the guard budget $K_{\max} = 24$, or on the patience stall.

Discount:
$\gamma = 0.99$.

3.2 The boundary-inward sweep prior

The project brief proposes an explicit placement order: "go around the edges and come inwards". This is a good prior for the AGP, because occlusion in a polygon with holes is generated at the boundary — specifically at reflex vertices, which are what make Chvátal's $\lfloor n/3 \rfloor$ bound tight [4, 7, 12] — and a guard placed against a wall sees a half-plane-like region with no wasted coverage behind it, whereas an interior guard's visibility polygon is partly redundant with itself.

We encode it as the term r_{sweep} of Document 1, a bounded exponential kernel on the distance-to-boundary d_∂ whose target radius sweeps inward *within* an episode:

$$r_{\text{sweep}}(\xi, t) \;=\; \exp\!\left(-\frac{\big(d_\partial(\xi) - \mu_{\text{ring}}(t)\big)^2}{2\sigma_s^2}\right), \qquad \mu_{\text{ring}}(t) = \Big(\mu_0 + (\mu_1 - \mu_0)\tfrac{t}{K_{\max}}\Big) \cdot L,$$

with L the bounding-box diagonal, $\mu_0 = 0.02$, $\mu_1 = 0.25$, $\sigma_s = 0.05L$. At $t = 0$ the policy is paid for placing within 2% of the diagonal of a wall; by the end of the guard budget the preferred band has moved to 25%, i.e. well into the interior.

Two properties matter. First, the kernel is bounded in $(0, 1]$, so its weight *is* its maximum per-step contribution — the convention every tracking reward in Wang et al. [20] and Xu et al. [21] follows, and the reason a dozen terms can be balanced by hand. Second, it is a prior on the placement *order*, not on the final configuration, which is structurally the same thing as the gait-phase indicators $\mathbf{1}_{\text{feet swing}} \cdot \mathbf{1}_{\text{swing period}}$ that the striker paper uses to install a walking rhythm in stage 1 and then zeroes in stage 2.

> r_{sweep} is *not* potential-based and therefore biases the optimum (Document 1, *anneal-iff-non-potential*): a wall-hugging guard set is not in general optimal. Its weight must reach 0 before SATURATE ends. We ramp it down linearly over the final 30% of the stage, so that the last part of SATURATE is trained under the unbiased objective and the policy has time to unlearn the prior's distortions before the gate is evaluated.

The same applies to r_{front} (pay for landing in still-uncovered space) and r_{sep} (punish placing within $d_{\min}$ of an existing guard). Both are useful early — they are what stop a random policy from stacking guards on one spot — and both are wrong late: in a narrow corridor two guards genuinely should sit close together, and a guard placed in already-covered space can still be the best next move if it opens a new region. They anneal on the same schedule.

3.3 Stage I-b: deliberate over-saturation

For the final ∼150k steps of SATURATE we change one thing: the episode no longer terminates at $c^\dagger$, and instead runs to $K_{\text{sat}} = \min\big(K_{\max}, \lceil 1.4\, k_{\text{ref}} \rceil\big)$ guards. The purpose is to manufacture the state distribution that PRUNE will be trained on. Without this, the policy hands PRUNE a guard set that is already close to minimal, almost no deletion is free, and the delete head learns nothing.

This is the AGP version of a trick both reference papers use in different guises: make the signal you want the agent to learn from artificially abundant early. Wang et al. [20] widen the simulated camera FOV to *twice* the real RealSense D455 field of view during stage 1, so that the active-sensing reward fires densely, then shrink it to the true value in stage 2. We inflate the guard count so that the redundancy signal fires densely, then let PRUNE deflate it.

By the *redundancy-monotonicity proposition* of Document 1, the redundancy deficits δ_j^- measured at K_{sat} guards are the smallest they will ever be, so this is also the state in which the pruning decision is least ambiguous.

3.4 Exit gate G_1

See section 7. Informally: held-out feasibility ≥ 0.90 on easy/mid and ≥ 0.75 on hard/extreme, sustained over three consecutive evaluation rounds, with all scaffolding weights already at zero.

4 Stage II — Prune

4.1 What changes

Three things, all of them in the weight column and the action mask:

(a) the mode head is unmasked to $\{\textsc{Add}, \textsc{Del}, \textsc{Stop}\}$;

(b) the guard-count potential is switched on, $w_{\text{par}} : 0 \to 0.02$, which by construction (Document 1) means -0.02 per Add and $+0.02$ per Del — exactly the "adding is negative, removing is positive" structure asked for;

(c) the clean-prune bonus $w_{\text{prune}} \exp(-\delta^{-}/\sigma_p)$ is switched on, which pays close to the full 0.15 when a deletion costs no coverage and decays to nothing as coverage is lost — the "lower reward scaling with space lost" clause.

Simultaneously all three scaffolding weights are already at zero, and the terminal parsimony bonus $w_{\text{pars}}(k_{\text{ref}} - k_T)$ turns on at 0.25. The coverage potential does *not* change: it is potential-based, so leaving it at $w_{\text{cov}} = 1.0$ costs nothing in bias and buys full protection against forgetting how to cover. This is where our design departs from Wang et al. [20] and Xu et al. [21], both of whom must demote their stage-1 proxy ($2.0 \to 1.0$ and $2.0 \to 0.5$) precisely because theirs is *not* potential-based and would otherwise compete with the real objective.

4.2 The delete action will not be discovered by accident

This is the main practical risk of the whole curriculum, and it has no counterpart in the reference papers, whose action space never changes. At the moment Del is unmasked, the policy has taken millions of Add actions and zero Del actions. Its mode-head logits for Del and Stop have received no meaningful gradient. Worse, the first deletions it tries will be uniformly distributed over guards and will therefore usually be expensive, producing negative advantage and pushing the Del logit further down. The likely outcome of a naive switch is a policy that never deletes and Prune that does nothing.

Four mitigations, in increasing order of effort.

1. Reset the mode head. Re-initialise the three mode logits to zero (uniform) at the switch, keeping the shared trunk and the point head. One line, and it removes the accumulated bias against unsampled modes.

2. A mode-specific entropy floor. Apply the entropy bonus to the categorical mode distribution separately from the continuous point distribution, with $\beta_{\mathcal{H}}^{\text{mode}} = 0.01$ decaying to 0.002 over the first 300k steps of Prune and $\beta_{\mathcal{H}}^{\text{point}} = 0.003$ throughout. The current code runs `ENT_COEF = 0.0`, which is fine for a single continuous head but wrong for a discrete head that must explore a newly available option.

3. A decaying logit bias, not forced actions. It is tempting to force Del with some probability ε during early Prune. Do not: PPO's importance ratio $\pi_\theta(a|s)/\pi_{\theta_{\text{old}}}(a|s)$ is computed against the policy that is assumed to have generated the data, and injecting off-policy actions silently corrupts it. Instead add a bias $b(t)$ to the Del logit *inside* the policy, so the behaviour policy and the policy being scored are the same distribution, and anneal b from 1.5 to 0 over 200k steps.

4. A privileged prune oracle with DAgger warm-start. The strongest option, and cheap here in a way it is not in robotics. Define:

Algorithm 1 GreedyPruneOracle(G, $\mathcal{G}$, $c^\dagger$)

```
while true do
    δ_j^- ← f(G) − f(G ⊖ g_j)  for all  g_j ∈ G
    j* ← arg min_j δ_j^-
    if f(G) − δ_{j*}^- < c† then
        break
    end if
    G ← G ⊖ g_{j*};  emit (Del, g_{j*})
end while
emit (Stop, ·)
```

One oracle call costs $O(k^2)$ Shapely unions, which for $k \leq 24$ is milliseconds — the per-guard visibility polygons are already cached in `Security.all_guards`. The oracle is *privileged* in the same sense as the teacher policies in Xu et al. [21] and Pinto et al. [13]: it uses exact continuous geometry that the rasterised student observation does not expose.

Use it exactly as they use their teacher, with DAgger [14]: roll out the student, query the oracle for the correct action at each visited state, and add a behaviour-cloning loss on the mode head and the point head,

$$\mathcal{L}_{\text{BC}} \;=\; \mathbb{E}_{s\sim\pi_\theta}\Big[\,\text{CE}(\mu_\theta(s),\,\mu^{\text{oracle}}(s)) + \|\xi_\theta(s) - \xi^{\text{oracle}}(s)\|^2\,\Big],$$

with coefficient 1.0 for the first 200k steps of Prune and linearly to 0 by 500k. Rolling out the *student* and labelling with the oracle (rather than cloning oracle trajectories) is the whole point of DAgger: the student is corrected on the states its own mistakes produce. Xu et al. [21] state the same rationale — "the student learns not only from the teacher's demonstrations but also from states encountered due to its own errors".

The oracle is a *warm start*, not the target. Greedy pruning is myopic: it deletes the cheapest guard at each step and never considers deleting two guards and re-adding one better-placed guard. The policy has both Del and Add available and can learn exactly that move, which is the main source of expected improvement over greedy. This is why $\mathcal{L}_{\text{BC}}$ must decay to zero — the same reason Wang et al. [20] delete their imitation reward entirely rather than decay it, "to promote exploration and discovery of effective dribbling behaviors".

4.3 Preventing coverage regression

The coverage ratchet of Document 1, $c_t \geq \min(\bar{c}_t, c^\dagger) - \delta_{\text{slack}}$, is enforced in Prune as an action mask: a Del that would violate it is rejected and charged r_{void}. Note the ratchet is measured against the *running maximum* $\bar{c}_t$, not against the previous step, so the 1% slack cannot be spent repeatedly — there is no drift exploit where the agent deletes a guard costing 0.9% twenty times over.

4.4 Episode initialisation

Prune episodes are drawn from a mixture:

Share	Initial state	Purpose
60%	empty gallery	the policy must still be able to solve an instance end to end
40%	saturated: K_{sat} guards from the frozen Saturate policy	dense pruning signal from step 0

The 40% saturated share is the direct analogue of moving the ball spawn from 10 m to 2 m [20] or tightening the robot spawn annulus from $[0.5, 4.0]$ to $[0.5, 2.0]$ m [21]: start the agent where the skill being learned is actually exercised. The 60% empty share is the insurance those papers buy differently, by demoting rather than deleting their stage-1 proxy.

4.5 Exit gate G_2

Feasibility not degraded by more than 2 percentage points relative to the end of SATURATE, mean pruning yield ≥ 2 guards, and competitive ratio $\mathbb{E}[k_T/k_{\text{ref}}] \leq 1.0$ — i.e. the policy is at least matching greedy on guard count while holding coverage.

5 Stage III — Refine (optional)

REFINE is worth running only if PRUNE converges and the residual failures are concentrated on the `hard`/`extreme` tiers, which is the expected outcome. It makes three changes.

5.1 From penalties to state-wise constraints

The regularity terms (r_{proj}, r_{void}, r_{stop}) move out of the reward and into a constraint,

$$\max_{\pi} \ \mathbb{E}\Big[\textstyle\sum_t \gamma^t r_t^{\text{task}}\Big] \quad \text{s.t.} \quad c^{\text{cost}}(s_t, a_t) \leq h \quad \forall (s_t, a_t) \in \tau,$$

solved with a constrained policy-gradient method such as N-P3O [9] or CPO [1]; see also Altman [2]. The motivation transfers exactly from Xu et al. [21], who diagnose *inhomogeneous credit assignment*: their task reward is concentrated at the kick, so "a fixed regularization coefficient can impose disproportionately large penalties at certain steps — particularly those immediately preceding the kick, where the agent expects high immediate future reward."

AGP has the identical profile. The first guard in an empty gallery typically captures 30–50% of the area; the last one before $c^\dagger$ captures a sliver. A fixed -0.25 aiming penalty is negligible against the first and dominant against the last, so it suppresses exactly the delicate end-game placements that determine whether the instance is solved at all. Their ablation is the reason to take this seriously: raising the fixed coefficient from 1.0 to 1.5 *dropped* success from 64.8% to 43.4%, while the constrained formulation reached 79.5%. That is the signature of a problem scalar tuning cannot fix.

Following their calibration heuristic — threshold set to "1.5× per-step average regularization cost of a walking gait" — we set

$$h \ = \ 1.5 \times (\text{mean per-step regularity cost of the greedy reference solver}),$$

i.e. calibrated against a known-good reference behaviour rather than in absolute units.

5.2 A relocation action

Add a fourth mode, MOVE: relocate the guard nearest to $\xi(a)$ by a bounded displacement emitted by a second continuous head. Pruning can only remove guards that are *already* redundant; relocation can *make* a guard redundant by improving a neighbour. This is the move that greedy fundamentally cannot make and is the main remaining source of improvement. It is listed as optional because it doubles the continuous action dimension and should not be introduced until PRUNE is stable.

Table 2: Reward weights by stage. "→" denotes a linear ramp within the stage; "—" denotes a term whose triggering action is masked. Terms in the scaffolding group are the only ones that *must* be scheduled (Document 1, *anneal-iff-non-potential*); every other schedule entry is an optimisation choice, not a correctness requirement.

Symbol	Group	Term	Saturate	Prune	Refine
w_{cov}	coverage	coverage potential (potential-based)	1.0	1.0	1.0
w_{goal}	coverage	terminal: $c_T \geq c^\dagger$	4.0	4.0	4.0
w_{fail}	coverage	terminal: $c_T < c^\dagger$	2.0	2.0	2.0
w_{par}	parsimony	guard-count potential (potential-based)	**0.0**	**0.02**	0.02
w_{prune}	parsimony	$\exp(-\delta^-/\sigma_p)$ on DEL	—	0.15	0.15
w_{pars}	parsimony	terminal: $k_{\text{ref}} - k_T$	0.0	0.25	0.35
w_{front}	scaffold	landed in uncovered space	$0.05 \to 0$	0	0
w_{sweep}	scaffold	boundary-inward ring prior	$0.10 \to 0$	0	0
w_{sep}	scaffold	minimum guard separation	$0.10 \to 0$	0	0
w_{proj}	regularity	aimed outside $\mathcal{F}$	0.25	0.25	constraint
w_{void}	regularity	invalid / masked DEL	—	0.10	constraint
w_{stop}	regularity	STOP with $c_t < c^\dagger$	—	2.0	constraint
β_{cov}	advantage weight, coverage critic		1	1	1
β_{par}	advantage weight, parsimony critic		0 *	2	2
β_{reg}	advantage weight, regularity critic		1	1	—
Action modes available			ADD	ADD, DEL, STOP	+ MOVE
Discount γ			0.99	0.997	0.997
$\mathcal{L}_{\text{BC}}$ (prune oracle)			—	$1.0 \to 0$	0

* The parsimony critic is still *trained* in SATURATE on the parsimony reward stream; only its advantage weight is zero, so it contributes no policy gradient. See section 7.3.

5.3 Hard-instance concentration

$\mathcal{D}$ reweights to $(0.10, 0.20, 0.35, 0.35)$ over the four tiers, and instances are additionally sampled in proportion to their recent failure rate — a simple prioritised-instance scheme. This is the only place where we use adaptive difficulty; none of the three reference papers has a performance-gated difficulty curriculum, though it is standard elsewhere in legged locomotion [10, 15].

6 The weight schedule

Table 2 is the curriculum. It is deliberately formatted the way Wang et al. [20] (their Table I) and Xu et al. [21] (their Table II) format theirs: one row per reward term, one column per stage.

Three entries deserve emphasis because they are the ones a reader will want to change first.

- $w_{\text{par}} = 0.0$ **in Saturate.** Not 0.005, not "small". Both multi-stage reference papers hold their eventual task reward at exactly zero through stage 1, and the project brief asks for the same. A small nonzero value is the worst of both worlds: too small to shape behaviour, large enough to make "place no guards" locally attractive to a random policy.
- $w_{\text{cov}} = 1.0$ **in every stage.** Because it is potential-based, its weight is a pure scaling of

an unbiased signal. Nothing is gained by demoting it and the risk of forgetting how to cover is real.

- $\beta_{\text{par}} : 0 \to 2$**.** The 2:1 ratio is taken directly from Wang et al. [19]'s $A_{\text{total}} = 2A_{\text{goal}} + A_{\text{aux}}$. The parsimony advantage is small in magnitude but low-variance (the reward is a constant $\pm w_{\text{par}}$ plus a bounded bonus), so it tolerates upweighting better than the coverage advantage does.

7 Stage transitions

7.1 Performance gates

Every $E = 10$k steps the current policy is evaluated deterministically on the full held-out set, per difficulty tier. Let $\text{SR}^{(i)}_{\text{easy/mid}}$ denote the feasibility rate ($\mathbb{P}[c_T \geq c^\dagger]$) at evaluation round i, and similarly for hard/extreme. The gates are:

$$G_1 : \quad \begin{cases} \text{SR}^{(i)}_{\text{easy/mid}} \geq 0.90 \\ \text{SR}^{(i)}_{\text{hard/extreme}} \geq 0.75 \\ \mathbb{E}[c_T] \geq 0.97 \end{cases} \quad \text{for } M = 3 \text{ consecutive } i, \quad \text{and} \quad w_{\text{front}} = w_{\text{sweep}} = w_{\text{sep}} = 0.$$

$$G_2 : \quad \begin{cases} \text{SR}^{(i)} \geq \text{SR}^{\text{end of SATURATE}} - 0.02 \\ \mathbb{E}[k_{\text{peak}} - k_T] \geq 2 \\ \mathbb{E}[k_T / k_{\text{ref}}] \leq 1.0 \end{cases} \quad \text{(pruning yield) for } M = 3 \text{ consecutive } i.$$

A hard step cap (1.5M for SATURATE, 1.5M for PRUNE) forces the transition if a gate never fires, with the failure logged. Requiring $M = 3$ consecutive rounds rather than a single crossing is the difference between a gate and a noise trigger: with only 50 held-out instances, a success rate near 0.9 has a binomial standard error of $\sqrt{0.9 \cdot 0.1/50} \approx 4$ percentage points, so a single evaluation carries a 95% interval more than 16 points wide. Enlarging the held-out set to 200 instances would be a cheaper improvement than tuning M.

7.2 Blending, and the value-function shock

A hard weight switch changes the reward function, and therefore the return, and therefore every value target the critic has been fitted to. The critic's error spikes, the advantage estimates become garbage for several updates, and PPO — which trusts its advantages — takes a large, badly-directed step. This is a plausible contributor to the instability already logged for this project (approx_kl in the tens, clip_fraction ≈ 0.7–0.8, policy collapse after 150–200k steps), and it will get worse, not better, once a second stage exists. None of the three reference papers discusses it.

Two mitigations, used together:

(i) **Linear blending.** Interpolate every weight linearly over $N_{\text{blend}} = 50$k steps: $w(t) = (1 - \alpha)w^{\text{old}} + \alpha w^{\text{new}}$, $\alpha = t/N_{\text{blend}}$. The action mask cannot be blended, so it flips at $\alpha = 0$; the associated weights ramp from zero, which means early deletions are nearly free and the policy can explore them before they start costing.

(ii) **Tighter trust region across the boundary.** Halve `target_kl` and the learning rate for the duration of the blend window, then restore. PPO's KL early-stop [17] is the right tool here and it is already enabled in the current code (`TARGET_KL = 0.03`).

7.3 The shadow parsimony critic

Because the reward is grouped (Document 1, multi-critic advantage decomposition), each group has its own value head, and a head can be trained without being used. During SATURATE we compute the parsimony reward stream ($-w_{\text{par}}$ per ADD, plus the terminal parsimony bonus) with the *PRUNE* weights, regress V_{par} against it, and set $\beta_{\text{par}} = 0$ so it contributes nothing to the policy gradient. At the switch, β_{par} ramps to 2 over the blend window against a value head that is already fitted.

This costs one extra MLP head and no environment changes, and it removes the single largest source of the transition shock. It has no counterpart in the reference papers, but it is a direct consequence of adopting Wang et al. [19]'s multi-critic decomposition: once the critics are separate, a critic can be warmed up in advance of the stage that needs it.

8 Architecture

8.1 Networks

Component	Specification
Actor trunk	CNN over the $64 \times 64 \times 5$ actor image: 32@8×8/4, 64@4×4/2, 64@3×3/1, ReLU, flatten, FC 256. Scalar branch: MLP $(64, 64)$. Concatenate $\to 320$.
Mode head	Linear $320 \to 3$, categorical over $\{\text{ADD}, \text{DEL}, \text{STOP}\}$ (plus MOVE in REFINE); invalid modes masked with $-\infty$ logits.
Point head	Linear $320 \to 4$, softplus $+ 1$, parameterising two independent Beta(α, β) distributions on $[0, 1]$ for a_x, a_y.
Critic trunk	Separate CNN over the $64 \times 64 \times 6$ privileged image plus the full scalar set; MLP $(512, 256, 128)$ (wider than the actor, as in Wang et al. [20]: 768/256/128 critic vs. 512/256/128 actor).
Critic heads	Three linear heads $\to 1$: V_{cov}, V_{par}, V_{reg}.
Activation	ELU throughout, following Wang et al. [19].

8.2 Why a Beta distribution for the point head

The action is a point in $[0, 1]^2$ and the current implementation samples a Gaussian and clips. This is a poor fit for the AGP specifically, because good guard positions are frequently *at* the boundary of the action range — against a wall, in a corner, in the mouth of a corridor. A clipped Gaussian piles a point mass of probability at the clip boundary, so the log-probability of a boundary action is wrong, the entropy is wrong, and the policy gradient at the boundary is biased inward [3]. A Beta distribution has support exactly $[0, 1]$, is bias-free at the boundary, and can represent both a sharp interior mode and a boundary-concentrated one. Given that the SATURATE sweep prior is explicitly *encouraging* boundary placements, this is not a minor detail.

8.3 Symmetry

The AGP is equivariant under the dihedral group D_4 (rotations by multiples of 90° and reflections) acting on the rasterised gallery, and under similarity transforms in the continuous problem. We exploit this twice:

- **Augmentation:** each instance is presented under a uniformly sampled D_4 transform, which multiplies the effective dataset by 8 at zero cost.
- **Consistency loss:** following the mirror-symmetry loss of Wang et al. [19] (their Eqs. 5–6, coefficient 10),

$$\mathcal{L}_{\text{sym}} = \mathbb{E}\Big[\big\|\xi_\theta(o) - M^{-1}\xi_\theta(Mo)\big\|^2 + \mathrm{KL}\big(\mu_\theta(o) \,\|\, \mu_\theta(Mo)\big)\Big], \quad M \sim D_4,$$

with coefficient 1.0: the point head should be equivariant, the mode head invariant. They needed this because their policy collapsed onto always kicking with one leg; the AGP analogue is a policy that collapses onto, say, always sweeping clockwise from the top-left.

9 PPO hyperparameters

Table 3: PPO settings per stage. The SATURATE column is close to the current code, with two changes: vectorised environments and a nonzero entropy coefficient.

Parameter	Saturate	Prune	Refine	Note
Discount γ	0.99	0.997	0.997	terminal parsimony must survive $\sim$ 30 steps
GAE λ	0.95	0.95	0.95	[16]
Clip range ϵ	0.2	0.15	0.10	tightened as the policy gets better
`target_kl`	0.03	0.02	0.015	halved during blend windows
Learning rate	1×10^{-4}	5×10^{-5}	3×10^{-5}	constant; halved during blend windows
Epochs per update	5	5	4	as in current code
Entropy: mode head	—	$0.01 \to 0.002$	0.002	critical for DEL discovery (section 4.2)
Entropy: point head	0.005	0.003	0.001	current code uses 0.0
Parallel envs	16	16	16	`SubprocVecEnv`; currently 1
Steps per env per update	256	256	256	rollout of 4096 transitions
Minibatches per epoch	4	4	4	as in Wang et al. [19]
Max grad norm	0.5	0.5	0.5	
Budget (steps)	1.5M	1.5M	0.75M	gates may trigger earlier

The single most impactful line in table 3 is probably **parallel envs**. The current run uses one environment, so a 2048-step rollout is one long, highly autocorrelated trajectory through a handful of galleries; the resulting gradient is a poor estimate of the population gradient, and this is a textbook cause of the enormous `approx_kl` values (10–100 against a target of 0.01–0.05) already recorded. The reference papers run 4,096 [20] and 16,384 [19] environments; we cannot match that with Shapely on CPU, but 16 processes is straightforward and changes the variance by more than an order of magnitude. Profile the visibility-polygon computation first — it is the bottleneck, it is $O(n \log n)$ per guard with n up to $\sim$100 vertices including holes, and it is called once per step.

10 Instance curriculum and augmentation

The locomotion papers randomise physics (friction, mass, CoM, actuator gains, latency) to close a sim-to-real gap. We have no sim-to-real gap; our generalisation gap is across *gallery instances*, so the analogue of domain randomisation is instance randomisation.

Mechanism	Detail
Difficulty mixture	Tier sampling weights $(0.55, 0.30, 0.10, 0.05) \to$ uniform over SATURATE; uniform in PRUNE; $(0.10, 0.20, 0.35, 0.35)$ in REFINE.
D_4 augmentation	Uniform random rotation/reflection of each instance at reset.
Scale jitter	Uniform scale in $[0.8, 1.25]$; all reward terms are already scale-free (areas normalised by A, distances by the bounding-box diagonal).
Vertex jitter	Gaussian perturbation of outer vertices with $\sigma = 0.01L$, rejecting non-simple results; generates fresh instances from a finite dataset.
Prioritised sampling	REFINE only: sample instances with probability $\propto (1 +$ recent failure rate).

Vertex jitter deserves a note: with $N_{\text{train}} = 500$ galleries and 1.5M steps at ~ 10 steps per episode, each gallery is seen roughly 300 times, which is well into memorisation territory. Jitter is the cheapest defence, and it is closer in spirit to what the locomotion papers do (they never reuse an exact initial state) than a fixed dataset is.

11 Evaluation and ablations

11.1 Baselines

- **Random interior placement** — the floor.
- **Greedy max-coverage** over a candidate set (all vertices, reflex vertices, and a 32×32 interior grid), which defines k_{ref}. This is the baseline to beat, and it is a strong one [6, 8, 11].
- **Greedy + oracle prune** (section 4.2) — the honest baseline for PRUNE, since our policy has access to the same pruning move.
- **Exact IP** [5, 18] on the `easy` tier only, to establish the true optimum where it is computable.

Reporting against greedy alone is not enough: if PRUNE only recovers what a greedy prune would have given anyway, the learned policy has added nothing.

11.2 Ablations

Table 4: Ablation protocol. Each row isolates one design decision from this document or Document 1.

Ablation	Change	What it tests
A1 Single stage	$w_{\text{par}} = 0.02$ and DEL unmasked from step 0	The core claim. Wang et al. [20] see 86.7% → 33.0% from this change.
A2 No pruning	SATURATE only, run for the full 3M steps	Isolates PRUNE's contribution from extra training.
A3 Sparse reward	Terminal c_T and k_T only, no potentials	Whether the dense telescoping decomposition matters (Document 1, telescoping proposition).
A4 No annealing	Scaffolding weights held constant	Tests the anneal-iff-non-potential rule directly; predicts a wall-hugging, over-separated final policy.
A5 Single critic	One value head on the summed reward	Wang et al. [19]'s multi-critic claim.
A6 No oracle	PRUNE without $\mathcal{L}_{\text{BC}}$	Whether DEL is discovered unaided (section 4.2).
A7 Hard switch	$N_{\text{blend}} = 0$, no shadow critic	Measures the transition shock (section 7.3).
A8 No sweep prior	$w_{\text{sweep}} = 0$ throughout	Whether "edges first, then inwards" helps or only slows convergence.
A9 Gaussian head	Clipped Gaussian instead of Beta	Boundary-action bias [3].
A10 Fixed penalty	REFINE with fixed coefficients instead of constraints	Xu et al. [21]'s inhomogeneous-credit claim.

Metrics are those of Document 1: feasibility rate, mean coverage, guard count conditioned on success, competitive ratio k_T/k_{ref}, pruning yield, and residual redundancy. Report all of them per difficulty tier, over at least three seeds — none of the three reference papers reports seed variance, which is a weakness worth not inheriting.

12 Implementation roadmap

Ordered by dependency, against the existing code.

R1. Fix truncation bootstrapping. Set `info["TimeLimit.truncated"] = True` on every `truncated` return in `ArtGalleryEnv.step`. One line; fixes a systematic value-target bias (Document 1, truncation-bootstrapping remark).

R2. Add the terminal reward. w_{goal}, w_{fail} in `step()`; w_{pars} once k_{ref} exists.

R3. Precompute k_{ref}. Extend `generate_polygon_dataset.py` to run greedy max-coverage per instance and store `k_ref` in the JSON record. Also store the greedy guard positions — they are useful for the oracle and for previews.

R4. Vectorise. `SubprocVecEnv` with 16 workers. Profile the ray sweep in `security.py` first; cache per-guard visibility polygons (already done via `all_guards`) and avoid recomputing the union from scratch on ADD.

R5. Extend the observation to the five actor channels and the privileged critic channels of Document 1's observation specification. This is the largest change and gates everything downstream.

R6. Extend the action space to $\texttt{Dict}(\texttt{mode}: \texttt{Discrete(3)}, \texttt{point}: \texttt{Box}(0, 1, (2,)))$ with masking. Note that Stable-Baselines3's stock PPO does not support action masking or Dict action spaces; either use `sb3-contrib`'s `MaskablePPO` with a factored discrete space, or implement a custom `ActorCriticPolicy` with a composite distribution. Budget real time for this.

R7. Weight scheduler. A callback that owns the stage state machine: current stage, blend progress, gate evaluation, and the weight vector pushed to the env via `set_attr`. Keep the schedule in one declarative table so it is diffable.

R8. Multi-critic. Three value heads, grouped reward returned in `info["reward_groups"]`, advantage combination per Document 1's multi-critic decomposition.

R9. Prune oracle and DAgger loss. section 4.2. Implement the oracle first as a standalone evaluation baseline (it is useful on its own), then wire in the BC loss.

R10. Beta policy head, D_4 augmentation, symmetry loss.

R11. Refine only after PRUNE is stable.

Items R1–R4 are worth doing regardless of whether the curriculum is adopted; they are straightforward fixes to the existing single-stage setup and several of them plausibly address the instability already observed.

13 Risks

The delete action is never used.
The main risk; section 4.2 gives four independent mitigations. Diagnostic: log the empirical mode distribution per stage. If $\mathbb{P}[\text{DEL}] < 0.05$ after 300k PRUNE steps, the warm start has failed.

Catastrophic forgetting of placement.
Mitigated by keeping w_{cov} fixed and by the 60% empty-gallery share in PRUNE. Diagnostic: feasibility rate must not fall by more than 2 points across the boundary — this is built into gate G_2.

Pruning that quietly costs coverage.
The ratchet against the running maximum $\bar{c}_t$ prevents drift, but $\epsilon^\star \approx 2.2\%$ (Document 1) still allows a single meaningful concession. If evaluation shows coverage creeping down, lower w_{par} rather than raising w_{prune}: the exchange rate $\epsilon = w_{\text{par}}/w_{\text{cov}}$ is what sets the tolerance.

Compute.
Every step calls exact ray-sweep visibility, and DEL additionally requires k set-difference evaluations for the redundancy channel. With 16 envs $\times$ 3M steps this is the dominant cost. Consider computing δ_j^- on a coarse raster for the observation while keeping the exact value for the reward.

Over-fitting to 500 galleries.
Addressed by vertex jitter and D_4 augmentation (section 10); monitor the train/held-out feasibility gap explicitly.

The curriculum is not needed.
Possible: Wang et al. [19] solve a comparably hard task *single-stage*, and attribute it to potential-based shaping plus a multi-critic split — both of which we also use. Ablation A1 is the test, and it should be run early rather than at the end, because a negative result would save most of this work.

14 Summary

1. **Saturate (Saturate)**: cover the gallery with the guard cost at exactly zero, under a boundary-inward placement prior and two other scaffolding terms that are all annealed to zero before the stage ends. A short over-saturation phase at the end deliberately manufactures the redundant guard sets that PRUNE learns from.
2. **Prune (Prune)**: unmask DEL and STOP, switch on the guard cost (so adding costs and removing refunds), and add a bounded bonus $\exp(-\delta^-/\sigma_p)$ that pays in full for a free removal and decays as coverage is lost. A coverage ratchet against the running maximum prevents regression.
3. **Refine (Refine, optional)**: move the regularity penalties into state-wise constraints, add a relocation action, and concentrate on hard instances.
4. The curriculum is entirely a weight schedule (table 2) plus an action mask over one fixed MDP. Only the three scaffolding terms *must* be annealed; everything else is potential-based and unbiased (Document 1).
5. Three things the reference papers do not supply and we add: quantitative stage gates, a blended transition with a shadow-trained parsimony critic, and a privileged greedy prune oracle used as a DAgger warm start for the newly unmasked delete action.
6. Run ablation A1 (single-stage) early. It is the one experiment that could invalidate the premise.

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